

Reading the Reader

Adaptive Reading Systems: A Modular Software Architecture

Master Thesis



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Date, year

By

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Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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1 Introduction

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1.1 Key Concepts and Related Works

1.1.1 Key Concepts for Adaptive Reading Analytics

This section defines the core concepts underlying the thesis. Definitions are grounded in primary and project-aligned sources.

Adaptive Reading System

The “Reading the Reader” concept assumes a closed-loop system that senses gaze-related signals (including pupil size) during reading, extracts higher-level events (fixations, saccades, regressions), classifies reading state or style, and triggers typographic interventions in real time [ref1].

Micro-Interventions and Just-in-Time Adaptation

The project proposal defines “micro-interventions” as small, context-preserving changes to typography or layout (e.g., spacing or contrast adjustments) designed to support reading without disrupting flow [ref2].

Within broader HCI literature, microintervention systems are described as structured, designed small actions embedded into systems, where timing, narrative coherence, and personalization influence overall user experience [ref2].

In this thesis, just-in-time adaptation is defined operationally as an intervention policy that triggers based on short temporal windows of online gaze and interaction features, while controlling for:

- Intervention cost (risk of disruption),
- Recovery behavior (time required to resume reading).

Context Preservation and Recovery Metrics

Recent ETRA 2025 work frames context preservation as interface functionality that enables readers to resume reading quickly after typographic changes. It introduces the metric Reading Resume Time (RRT) and compares four intervention designs: Popup, Undo, Notification, and Gradual [ref2].

Results indicate that a Gradual intervention combined with highlighting yields the fastest recovery time and highest user preference. This demonstrates that intervention form factor and transition smoothness significantly affect usability.

In this thesis, context preservation is treated as a first-class design goal with measurable outcomes:

- Reading Resume Time (RRT),
- Gaze stabilization time,
- Regression bursts following change.

1.1.2 Related Academic Work

Context Preservation and Intervention Design

The ETRA 2025 paper on context preservation formalizes a key HCI challenge: adaptive reading interfaces can either assist or hinder reading depending on how changes are introduced and how recovery is supported. In a within-subject study with 22 participants, significant differences in RRT were reported across intervention types [ref2].

This supports the thesis argument that intervention design is a measurable system behavior that must be architecturally supported through stable integration hooks and logging mechanisms.

Reading Analytics for Emerging Content Types

An ETRA 2025 study comparing AI-generated and human-authored texts collected gaze data from 13 participants and applied I2MC for fixation identification. The study also analyzed pupil dilation and reading speed [ref2].

Findings demonstrate:

1. Gaze metrics are sensitive to subtle content differences,
2. Experimental reproducibility requires modular and transparent data pipelines.

1.1.3 Market Landscape of Tools and Products

Comparison of Tools

Category	Vendor / Product		Delivery Model	Capabilities	Pricing Signal
Screen-based eye tracker	Tobii Pro Fusion		Hardware	Up to 250 Hz sampling; fixation/saccade detection; pupil diameter recording [ref3]	Quote-based [ref3]
Screen-based eye tracker	EyeLink Plus	1000	Hardware	Binocular sampling up to 2000 Hz; research integration support [ref4]	Quote request [ref5]
Lower-cost eye tracker	GazePoint HD	GP3	Hardware + Software	150 Hz sampling; research-grade positioning [ref6]	\$2,650 hardware-only [ref6]
Wearable eye tracking	Pupil Labs Neon		Hardware	Academic pricing; complete research system [ref7]	€5,515 academic [ref7]
Remote webcam tracking	RealEye		SaaS	Subscription plans; data export; panel workflow [ref8]	Panelists \$15 [ref9]
Experiment software	Tobii Pro Lab		Desktop software	Experiment design + recording + analysis [ref10]	Trial available [ref10]
Reading assessment	Lexplore		Service + Hardware	AI-based reading assessment [ref11]	Quote-based [ref12]
Accessibility device	OrCam Read 3		Hardware	Reads printed/digital text aloud [ref13]	\$116/month option [ref13]
Adaptive interface	Microsoft Immersive Reader		Built-in feature	Line Focus; Text Spacing; read-aloud [ref14]	Included [ref14]
E-reading customization	Amazon Kindle settings		Device feature	Font size and appearance adjustments [ref15]	Included with device [ref15]

1.1.4 Gaps and Opportunities

Architecture Gap: Experimentability

The project proposal highlights that tightly coupled prototypes hinder systematic comparison of interventions and decision strategies. Empirical findings show that intervention design materially affects recovery and user preference [ref2, ref16].

Opportunity: Design the adaptation engine with hot-swappable intervention modules and auditable logging to enable:

- Systematic A/B testing,
- Replay-based evaluation,
- Real-time override capabilities,
- Transparent experimental comparison.

1.1.5 Conclusion

The reviewed literature and market landscape collectively demonstrate that adaptive reading systems require:

1. Robust gaze-processing pipelines,
2. Carefully designed intervention mechanisms,
3. Context-preservation metrics,
4. Modular, experiment-ready architectures.

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A Title

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